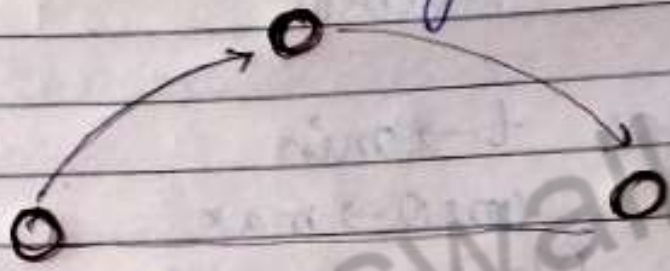
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PROJECTILE MOTION

* Actual Meaning of projectile motion is motion under gravity.

* And in syllabus it is Motion in a plane (2-D)

eg:- When a ball is thrown upward with some angle.



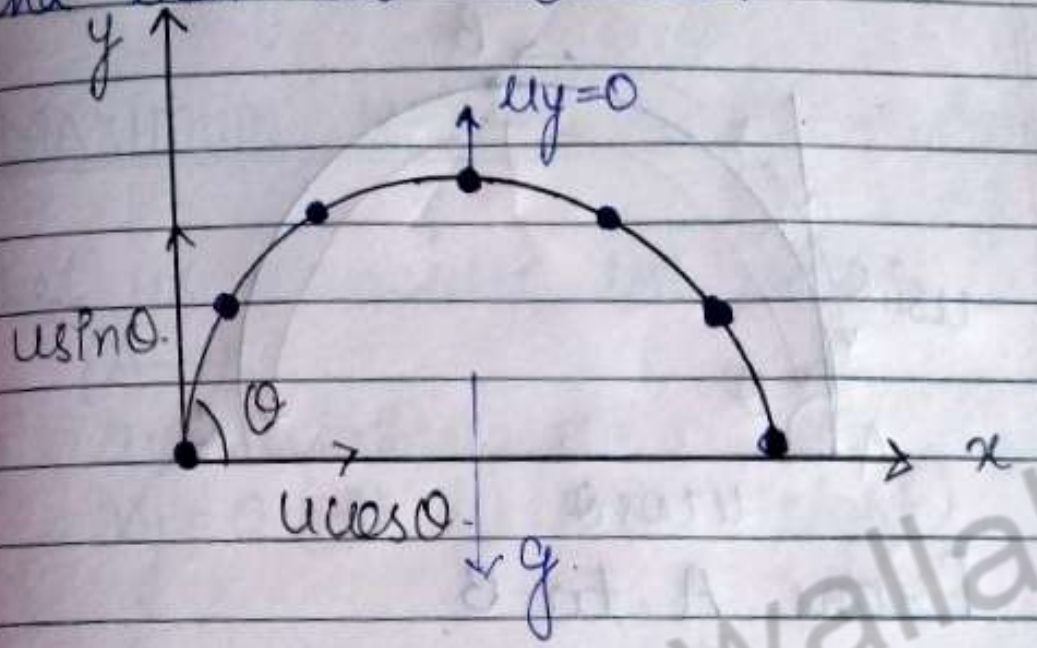
In a projectile motion,
u is the initial velocity,
θ is the angle of projection,
the motion of an object is in 2 direction, so
Initial velocity in x direction is given by $u_x = u\cos\theta$,
Initial velocity of y direction is given by $u_y = u\sin\theta$.

⇒ Acc. due to gravity in x-direction.
$a_x = 0$

⇒ Acc. due to gravity in y-direction
$a_y = -g$.

"Whenever you take a step forward, you are bound to disturb something." —Indira Gandhi

so, x direction velocity is always constant. whereas, velocity of y direction first increases, and at highest point it becomes 0 and then it increases.



Assuming there is no air resistance, and no friction

In Projectile Motion

⇒ Time of flight is given by T
⇒ Max. Height is given by H
⇒ Horizontal Range is given by R

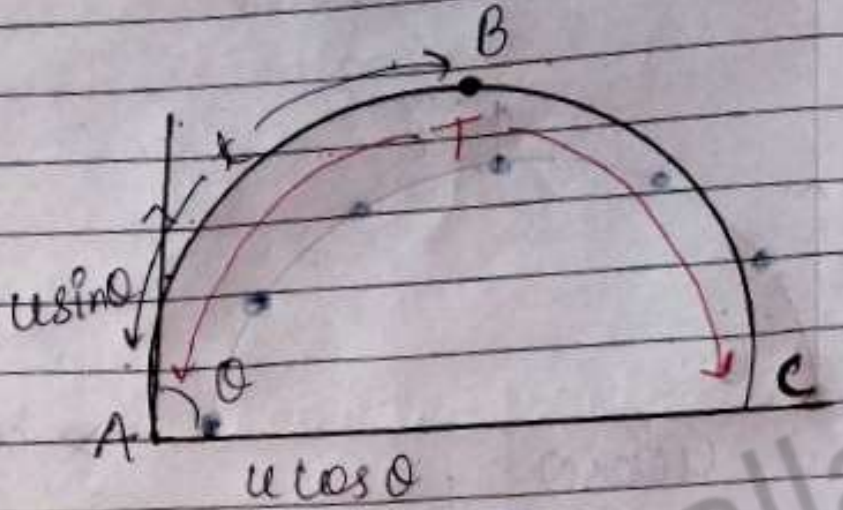
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Time of flight $\left(T = \frac{2u\sin\theta}{g}\right)$

Let us consider motion in y-direction



From A to B

$u_y = u\sin\theta$,

& we know $a_y = -g$, and

$v_y = 0$ (at B)

Let the object took time (t)

From first equation of motion

$$v = u + at$$

$$0 = u\sin\theta + (-g) \times t$$

$$\left[t = \frac{u\sin\theta}{g}\right] \text{ upto B}$$

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"Whenever you take a step forward, you are bound to

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Total time = T

$T = 2t$

$$\boxed{T = \frac{2u\sin\theta}{g}}$$ Time of flight

MAXIMUM HEIGHT $\left(H = \frac{u^2\sin^2\theta}{2g}\right)$

Let us consider the motion in y direction (A → B)

$u_y = u\sin\theta$

$v_y = 0$ (At B) (Max Height)

$a_y = -g$

Displacement in Y-direction = H

Using 3rd equation of motion

$v^2 = u^2 + 2as$

$0^2 = (u\sin\theta)^2 + 2(-g)H$

$2gH = u^2\sin^2\theta$

$$\boxed{H = \frac{u^2\sin^2\theta}{2g}}$$

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HORIZONTAL RANGE $\left(R = \frac{u^2 \sin 2\theta}{g}\right)$

Let us consider the motion in x-direction
$(A \to C)$

$u_x = u\cos\theta$

$a_x = 0$

$s_x = R$

$t = T$

Using second equation of motion

$$s = ut + \frac{1}{2}at^2$$

$$R = u\cos\theta \times \frac{2u\sin\theta}{g}$$

$$\boxed{R = \frac{u^2 2\sin\theta\cos\theta}{g}}$$

OR

$$\boxed{R = \frac{u^2 \sin 2\theta}{g}}$$

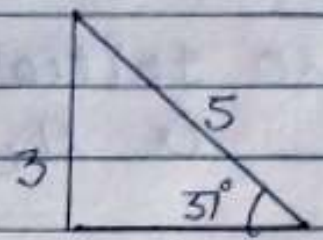
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Ques:- A ball is thrown with 5 m/sec at an angle of projection 37°. Find
(i) Time of flight.
(ii) Max. Height
(iii) Horizontal Range

Ans(i) $T = \frac{2u\sin\theta}{g}$

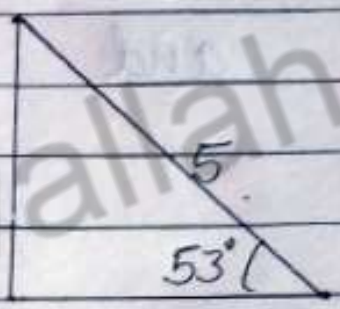
$\theta \to 37°$



$T = \frac{2 \times 5 \times \sin 37°}{10}$

$T = \frac{3}{5} = 0.6$ sec

(ii) $H = \frac{u^2 \sin^2\theta}{2g}$



$H = \frac{25(\sin 37°)^2}{2 \times 10}$

$H = \frac{9}{20}$

(iii) $\text{Range} = \frac{u^2 \sin 2\theta}{g} = \frac{u^2 2\sin\theta\cos\theta}{g}$

$R = \frac{12}{5} m = 2.4 m$

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"No religion has mandated killing others as a requirement for its sustenance or promotion." —Dr.A.P.J.Abdul Kalam

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Ques6- A ball is thrown with 50m/sec at angle of projection 37°. Find velocity vector and speed of particle after 2 sec. of projection.

Ans6- So, Initial velocity in x direction is $u_x = u\cos\theta$

$= 50 \times \frac{4}{5}$

$u_x = 40$ m/sec;

and $u_y = u\sin\theta$

$u_y = 30$ m/sec.

After 2 sec,

Velocity ~~Speed~~ of particle in x-direction is

$v_x = 40$ m/sec.

Velocity ~~Speed~~ of particle in y-direction is v_y

$u_y = 30$ m/sec.

$a_y = -10$

$t = 2$ sec.

$v_y = u_y + at$

$v_y = 30 + (-10)2$

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$v_y = 30 - 20$

$v_y = 10$ m/sec.

In vector form.

$$\vec{v} = 40\hat{j} + 10\hat{j}$$

Speed $|v| = \sqrt{(40)^2 + (10)^2}$

$$= \sqrt{1600 + 100}$$

$$|v| = \sqrt{1700}$$

Ques 8- For a projectile motion from ground to ground, $H = R$, find angle of projection

Ans:-

$$H = R$$

$$\frac{u^2 \sin^2\theta}{2g} = \frac{u^2 \sin 2\theta}{g}$$

$$\frac{\sin^2\theta}{2} = \sin 2\theta$$

$$\frac{\sin^2\theta}{2} = 2\sin\theta\cos\theta$$

$$\frac{\sin\theta}{2} = 2\cos\theta$$

$$\tan\theta = 4$$

$$\theta = \tan^{-1} 4.$$

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Ques8:- For a projectile Motion from ground to ground. If Max. Ht is $30\sqrt{3}$ m and Horizontal Range is 120m. Find Initial velocity and angle of projection (u, θ)

Ans:-

$H = 30\sqrt{3}$

$R = 120$

$$\frac{u^2 \sin^2\theta}{2g} = 30\sqrt{3} \quad \text{— (i)}$$

$$\frac{u^2 \sin 2\theta}{g} = 120 \quad \text{or} \quad \frac{2u^2 \sin\theta\cos\theta}{g} = 120 \quad \text{— (ii)}$$

Dividing eq (i) by (ii) we get

$$\frac{u^2 \sin^2\theta}{2g} \times \frac{g}{2u^2 \sin\theta\cos\theta} = \frac{30\sqrt{3}}{120}$$

$$\frac{\sin\theta}{4\cos\theta} = \frac{30\sqrt{3}}{120}$$

$$\tan\theta = \sqrt{3}.$$

$$\boxed{\theta = 60^\circ}$$

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$$\frac{u^2 \sin^2\theta}{2g} = 30\sqrt{3}$$

$$\frac{u^2 \sin^2 60^\circ}{2g} = 30\sqrt{3}$$

$$\frac{u^2\, 3}{2 \times 10 \times 4} = 30\sqrt{3}$$

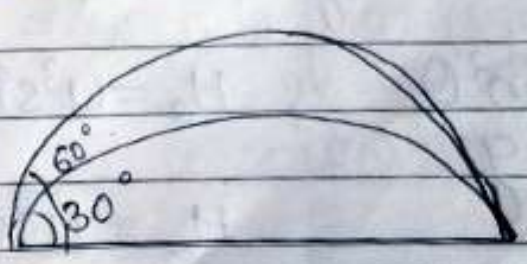
$$u^2 = \frac{80 \times 30\sqrt{3}}{3}$$

$$u^2 = \frac{2400}{\sqrt{3}}$$

$$u = \sqrt{\frac{2400}{\sqrt{3}}}$$

RANGE : Range is same for two different angles of projection if u is same

θ°, '90-θ', suppose angles are 30° and 60°.



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PROOF

$$\boxed{R = \frac{u^2 \sin 2\theta}{g}} \quad (\text{For } \theta)$$

For $(90-\theta)$

$$R = \frac{u^2 \sin 2(90-\theta)}{g}$$

$$R = \frac{u^2 \sin(180-2\theta)}{g}$$

$\sin(180-\theta) = \sin\theta$

$$\boxed{R = \frac{u^2 \sin 2\theta}{g}}$$

For two angle of projection, Range is same, and vertical max Heights are different ie H_1 & H_2. Find relation between H_1, H_2 & R.

For θ, Height is H_1
For $(90-\theta)$, Height is H_2

$$H_1 = \frac{u^2 \sin^2\theta}{2g}, \quad H_2 = \frac{u^2 \sin^2(90-\theta)}{2g}$$

$$H_2 = \frac{u^2 \cos^2\theta}{2g}$$

$u\sin\theta = \sqrt{2gH_1}$

$u\cos\theta = \sqrt{2gH_2}$

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"Whenever you take a step forward, you are bound to disturb something." - Indira Gandhi

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$$R = \frac{u^2 \sin 2\theta}{g} = \frac{u^2 2 \sin\theta \cos\theta}{g}$$

$$= \frac{(u\sin\theta)(u\cos\theta) \times 2}{g}$$

$$R = \frac{\sqrt{2gH_1}\,\sqrt{2gH_2} \times 2}{g}$$

$$R = 2\sqrt{4H_1H_2}$$

$$R = 4\sqrt{H_1H_2}$$

When Range will be Maximum.

$$R = \frac{u^2 \sin 2\theta}{g}$$

Range depends upon speed and θ.

If $\theta \to 45^\circ$
$R \to$ max.

when $\frac{dR}{d\theta} = 0$ (Range will be Maximum)

$$\frac{dR}{d\theta} = \frac{u^2 \cos 2\theta \times 2}{g}$$

$$\frac{u^2 \cos 2\theta \times 2}{g} = 0$$

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$\cos 2\theta = 0$

$2\theta = 90°$

$\theta = 45°$

$$R = \frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin 90}{g}$$

$$\boxed{R_{max} = \frac{u^2}{g}}$$

Max Range is $\frac{u^2}{g}$ when $\theta = 45°$

Path of projectile Motion is parabolic.

⇒ condition of parabola

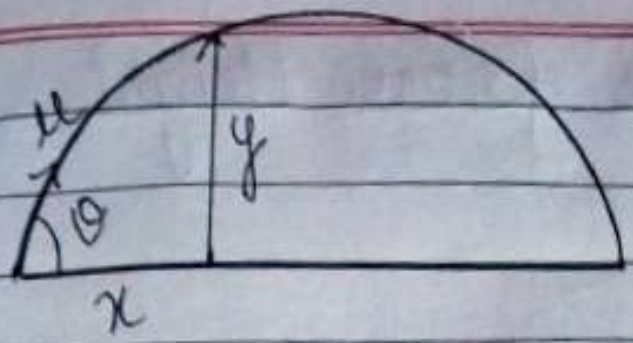
$$y^2 \propto x$$

$$\underline{y^2 = 4ax}$$

When in an equation y & x are relatable that equation is called as equation of trajectory (Path) time (t) is not included.

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x

$u_x = u\cos\theta$

$a_x = 0$

t

$s = ut + \frac{1}{2}at^2$

$x = u\cos\theta\, t$

$t = \frac{x}{u\cos\theta}$

y

$u_y = u\sin\theta$

$a_y = -g$

t

$s = ut + \frac{1}{2}at^2$

$y = u\sin\theta\, t + \frac{1}{2}(-g)t^2$

$$y = u\sin\theta\left(\frac{x}{u\cos\theta}\right) - \frac{1}{2}g\cdot\frac{x^2}{u^2\cos^2\theta}$$

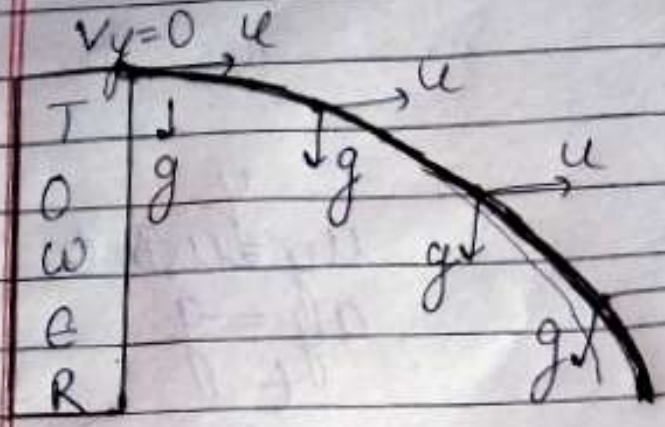
$$\boxed{y = x\tan\theta - \frac{1}{2}\frac{gx^2}{u^2\cos^2\theta}}$$ (Parabola)

This is called Equation of Trajectory.

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Projectile from Height

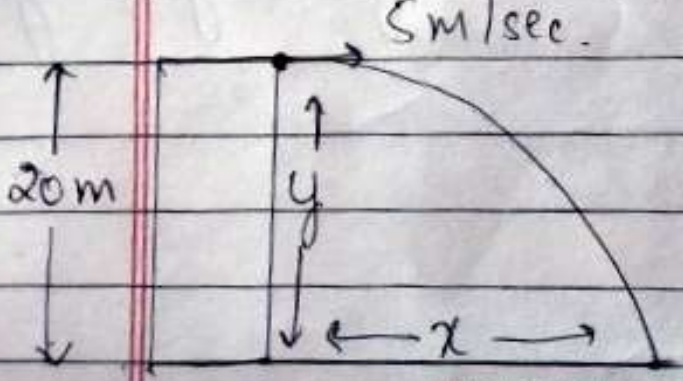


2-D motion

Velocity in x-direction remains constant whereas velocity in y direction increases (due to g)

Path → Parabolic path (projectile from height)

Ques- consider a building of 20m, from the top of building we threw a particle horizontally at the speed of 5m/sec. Find the time taken by particle to reach ground.



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Ans:- It is a 2-D motion (covers a distance in x & y direction).

Let us consider whole motion in y-direction.

$S_y = -20m$ (displacement)

$u_y = 0$, $a_y = -g = -10$

$s = ut + \frac{1}{2}at^2$

$-20 = 0 + \frac{1}{2}(-10)t^2$

$\boxed{t = 2 sec}$

⇒ Find the Horizontal range.

Let us consider motion in x-direction

$S_x = R$

$u_x = 5m/sec$

$a_x = 0$, $t = 2sec$.

$s = ut + \frac{1}{2}at^2$.

$R = 5 \times 2 + 0$.

$\boxed{R = 10m}$

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⇒ Find the velocity of ball when it hits the ground

$\vec{v} = \vec{v}_x + \vec{v}_y$

y-direction

$u_y = 0$

$a_y = -10$

$v_y = ?$

$v = u + at$

$v = -20 \text{ m/sec}$

$\vec{v} = \vec{v}_x + \vec{v}_y$

$\boxed{\vec{v} = 5\hat{i} - 20\hat{j}}$

⇒ Speed

Speed = Mag. of velocity

$\text{Speed} = \sqrt{(5)^2 + (-20)^2}$

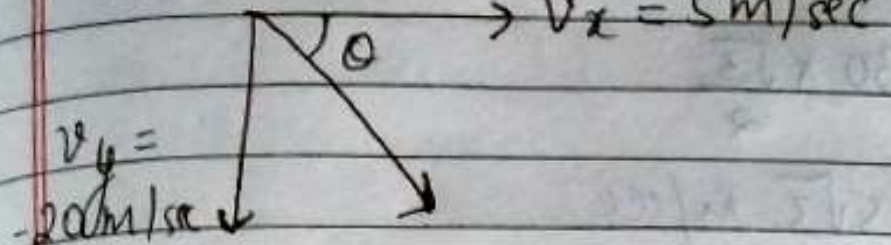
$\text{Speed} = \sqrt{25 + 400}$

$\text{Speed} = \sqrt{425}$

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⇒ Find θ.



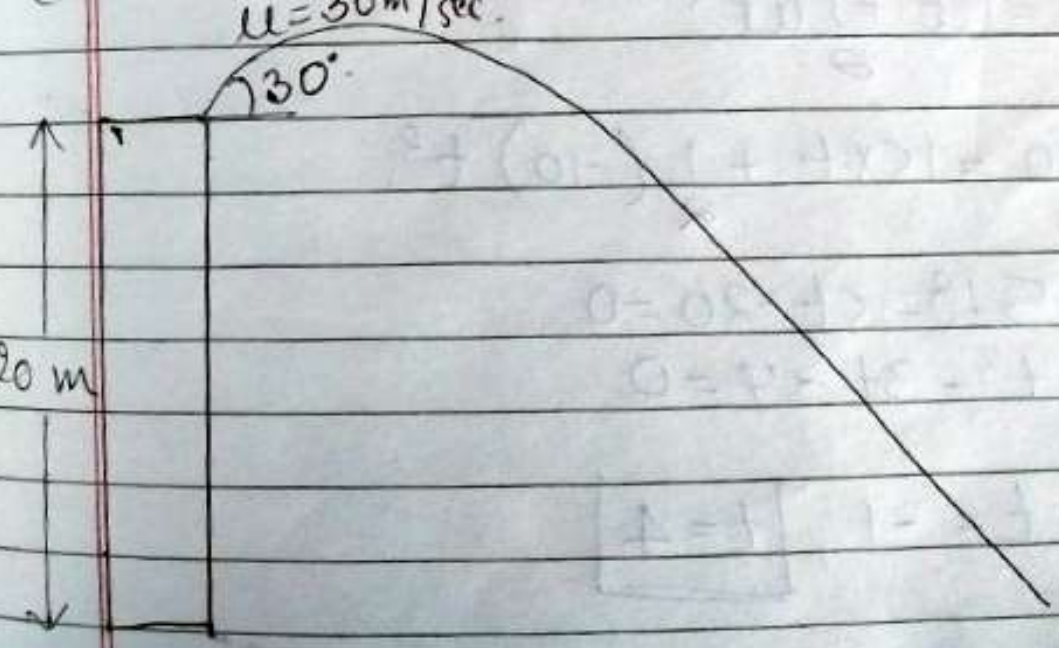
$$\tan\theta = \frac{v_y}{v_x}$$

$$\tan\theta = \frac{20}{5}$$

$$\tan\theta = 4$$

$$\theta = \tan^{-1}4$$

Ques- If a ball is thrown at 30° angle from a (20m) tower with initial speed 30m/sec.
(i) Find the time taken to reach ground.



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In x-direction

$u_x = u\cos 30^\circ$

$u_x = 30 \times \frac{\sqrt{3}}{2}$

$u_x = 15\sqrt{3}$ m/sec.

In y-direction

$u_y = u\sin\theta$.

$u_y = 15$ m/sec.

(i) Let us take motion in y-direction

$s_y = -20$ (shortest between initial and final position)

$a_y = -10$

$s = ut + \frac{1}{2}at^2$

$-20 = 15 \times t + \frac{1}{2}(-10)t^2$.

$5t^2 - 15t - 20 = 0$.

$t^2 - 3t - 4 = 0$.

$t = -1, \boxed{t = 4}$

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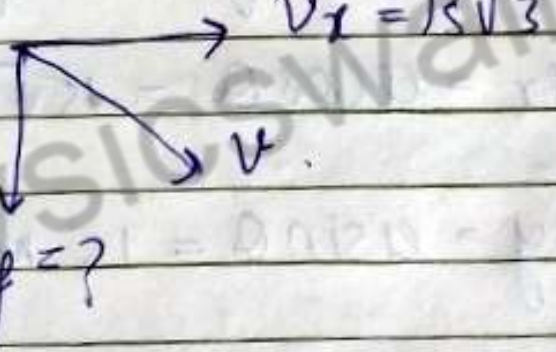
(ii) Find Horizontal range

$u_x = 15\sqrt{3}$.
$a_x = 0$
$t = 4$ sec.
$S_x = R$

$$S = ut + \frac{1}{2}at^2$$

$$\boxed{R = 60\sqrt{3}\ m}$$

(iii) final velocity just before hitting ground.



y-direction
$u_y = 15$
$a_y = -10$
$t = 4$ sec.
$v_y = ?$

$v = u + at$
$v = -25$ m/sec.

$$\boxed{v_y = -25\ m/sec}$$

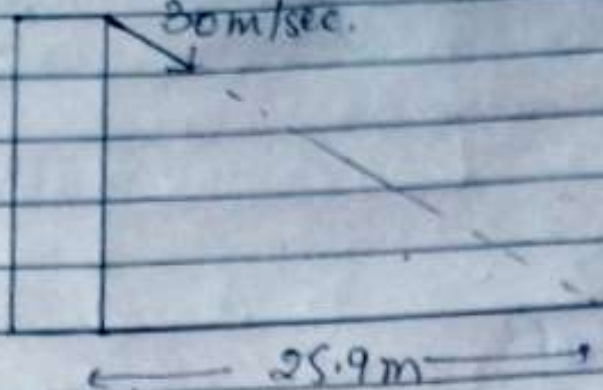
Velocity $= 15\sqrt{3}\hat{i} - 25\hat{j}$.

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Ques:-



Find the height of tower?

Ans:- Component of velocity.

$$u_x = u\cos\theta = 15\sqrt{3} \text{ m/sec}$$

$$u_y = u\sin\theta = 15 \text{ m/sec.}$$

X-direction Motion

$$u_x = 15\sqrt{3}$$

$$a = 0$$

$$S_x = 25.9$$

$$t = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$t = \frac{25.9}{15\sqrt{3}}$$

Y-direction Motion

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$$S = ut + \frac{1}{2}at^2$$

$$S = \frac{15 \times 25\sqrt{3}}{15\sqrt{3}} + \frac{1}{2}(-10) \times t^2$$

EQUATION OF TRAJECTORY.

$$\left\{ y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} \right\}$$

$$y = x\tan\theta - \frac{gx^2 \sin\theta}{2u^2\cos^2\theta \sin\theta}$$

$$y = x\tan\theta - \frac{gx^2\tan\theta}{u^2 2\sin\theta\cos\theta}$$

$$y = x\tan\theta - \frac{x^2\tan\theta}{\frac{u^2\sin 2\theta}{g}}$$

$$y = x\tan\theta - \frac{x^2\tan\theta}{R}$$

$$\left\{ y = x\tan\theta\left(1 - \frac{x}{R}\right) \right\}.$$

This is also equation of trajectory.

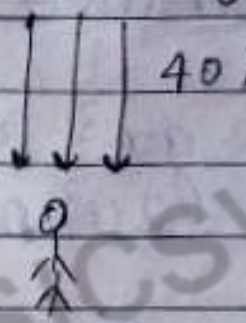
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Relative Velocity 2-D, Rain Man Problem, Umbrella Woman Problem.

$$\boxed{\vec{v}_{AB} = \vec{v}_{AG} - \vec{v}_{BG}} \quad 2\text{-D}$$

Ques- The speed of rain is 40 km/hr. A man is standing on ground. What is the speed of Rain as seen by man



Ans- $\vec{v}_{RM} = \vec{v}_{RG} - \vec{v}_{MG}$

$\vec{v}_{RM} = -40\hat{j} - 0$

$\vec{v}_{RM} = -40\hat{j}$ (The man will open its umbrella off to the direction of rain)

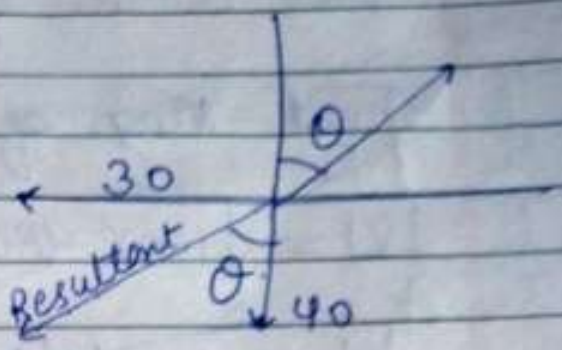
⇒ Now the man is in a cycle with speed 30 km/hr at what angle should he bend his umbrella from vertical.

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$v_{rm} = v_{rG} - v_{mG}$

$v_{rm} = -40\hat{j} - 30\hat{i}$

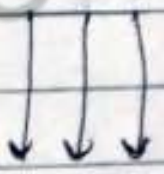


$\tan\theta = \frac{30}{40} = \frac{3}{4}$

$\theta = 37^\circ$.

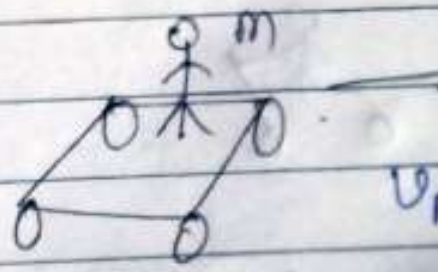
The man should hold the umberella 30° from vertical position

Ques 6- If the speed of rain is 60 km/hr vertically. A man is in trolley of speed 80 km/hr, At what angle with the man open's its umberella.



$R = 60$ km/hr

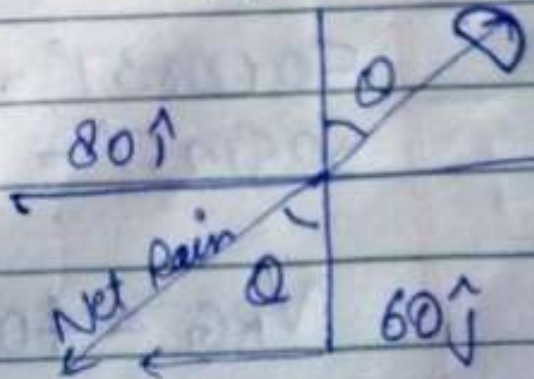
$v_{RG} = -60\hat{j}$



$v_{mG} = 80\hat{i}$

$v_{RM} = v_{RG} - v_{mG}$

$v_{RM} = -60\hat{j} - 80\hat{i}$



$\tan\theta = \frac{80}{60} = \frac{4}{3}$

$\theta = 53^\circ$

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⇒ What is the speed of rain w.r.t man

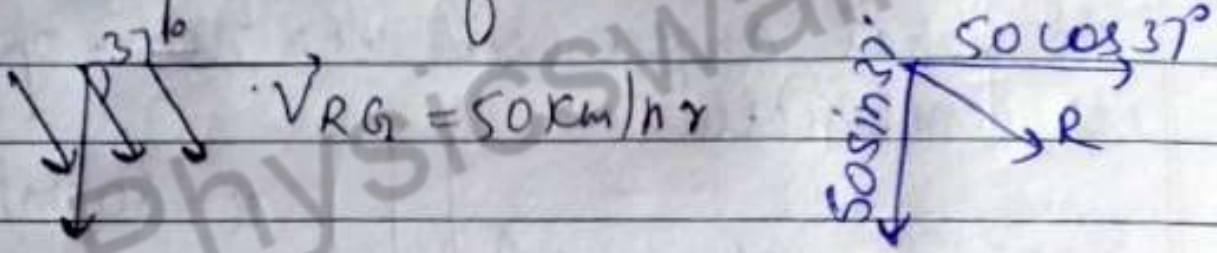
speed = Mag. of velocity

$|V| = \sqrt{(60)^2 + (80)^2}$

$|V| = 100$ ~~km/hr~~

$|V| = 100 \text{ km/hr}$

Ques6- If rain is at 37° angle & $V_{RG} = 50$ km/hr a man in cycle at a speed of 40 km/hr V_{MG} (+x direction). find the position of umbrella.





$50\cos 37° = 40$

$50\sin 37° = 30$.

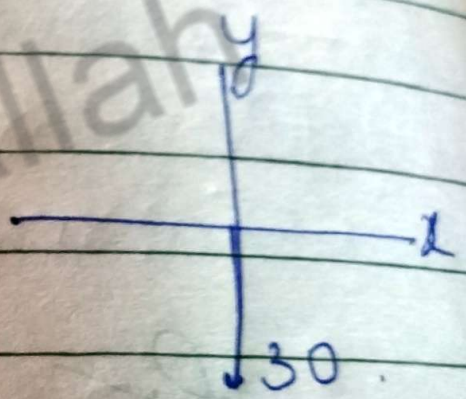
$V_{RG} = 40\hat{i} - 30\hat{j}$

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$$\vec{V}_{RM} = \vec{V}_{RG} - \vec{V}_{MG}$$

$$= 40\hat{j} - 30\hat{j} - 40\hat{j}$$

$$\vec{V}_{RM} = -30\hat{j}.$$



Umbrella should be vertically upward.

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